

SMI CRLs: the installed beamline

CURRENT BEST UNDERSTANDING

Reconstructing the operating beamline - September 2026

Original installed-system documentation is no longer available.

This records our current best understanding and calculated settings.

LAYOUT

Lens plane to sample at Z=0: 1600 mm

Z travel: -300 to 300 mm; positive downstream

SSA secondary source: 10.5 m upstream of CRL Z=0.

KNOWN INVENTORY (sizes are quoted radii)

Holder	Contents	IN mm	OUT mm
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1	1 x 50 um Be	0.0	10.0
2	8 x 50 um Be	0.0	10.0
3	16 x 50 um Be	0.0	10.0
4	4 x 50 um Be	0.0	10.0
5	2 x 50 um Be	0.0	10.0
6	blank / uncharacterized	0.0	10.0
7	blank / uncharacterized	0.0	-10.0
8	blank / uncharacterized	0.0	-10.0
9	1 x 200 um Be	0.0	-10.0
10	8 x 500 um Be	0.0	-10.0
11	1 x 500 um Be	0.0	-10.0
12	2 mm aperture	0.0	-10.0

Blank holders 6-8 may contain diamond CRLs; contents unconfirmed.
Locate and measure the 12 reported diamonds during commissioning.
Blanks and aperture 12 are excluded from automatic lens selection.
IN=0 mm is nominal, pending alignment within about +/-2 mm.

CALCULATION BASIS

Two-sided parabolic lenses assumed; all labeled lenses are Be.

Be density 1.848 g/cm³; free-electron delta scales as energy⁻².

Common-plane thin lenses image the SSA onto the sample.

Predicts geometrical focus, not spot size or transmission.

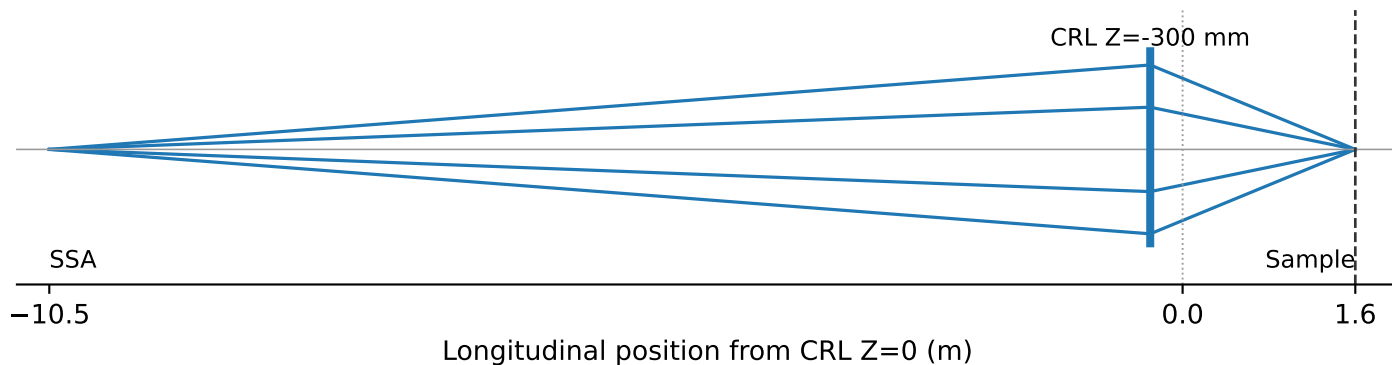
FOLLOW-UP

Test a few energies/combinations; confirm and measure blank contents (diamond?); develop advanced alignment routines.

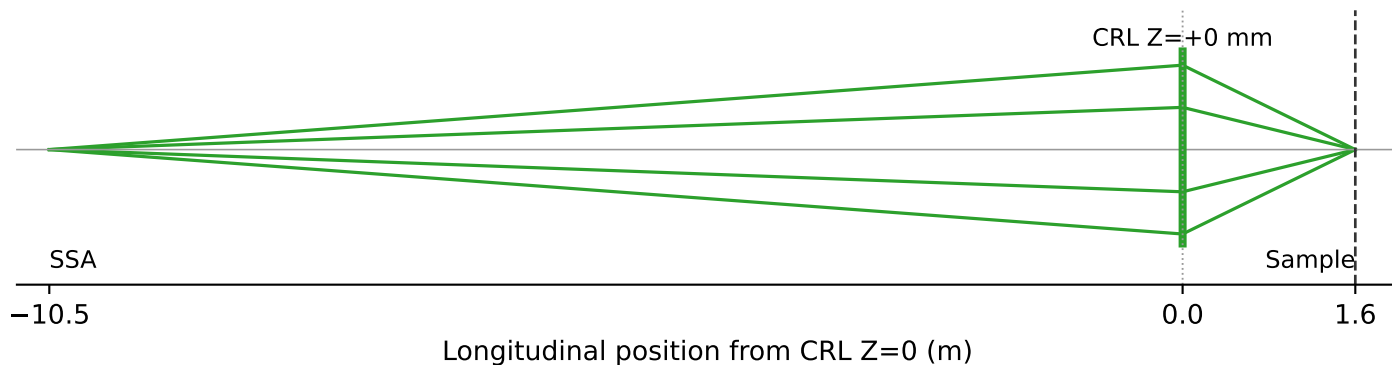
SSA to CRL to sample

Rays at the travel limits and center, focused onto the fixed sample.
Each panel uses the required lens power at that position; the holder-3 energy is an equivalent example, not the same energy in all three panels.

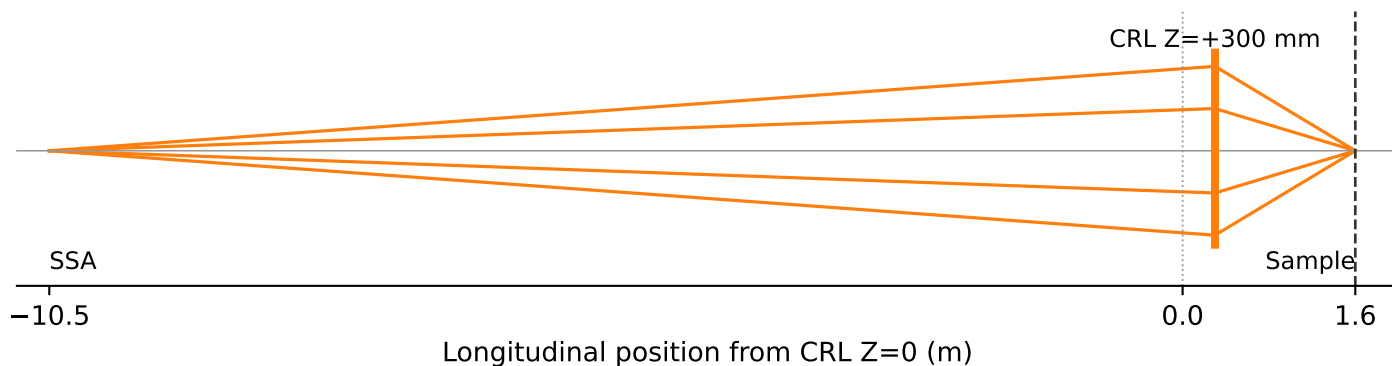
$p=10.200$ m; $q=1.900$ m; required $f=1601.7$ mm
Holder 3 example: 18.683 keV



$p=10.500$ m; $q=1.600$ m; required $f=1388.4$ mm
Holder 3 example: 17.395 keV

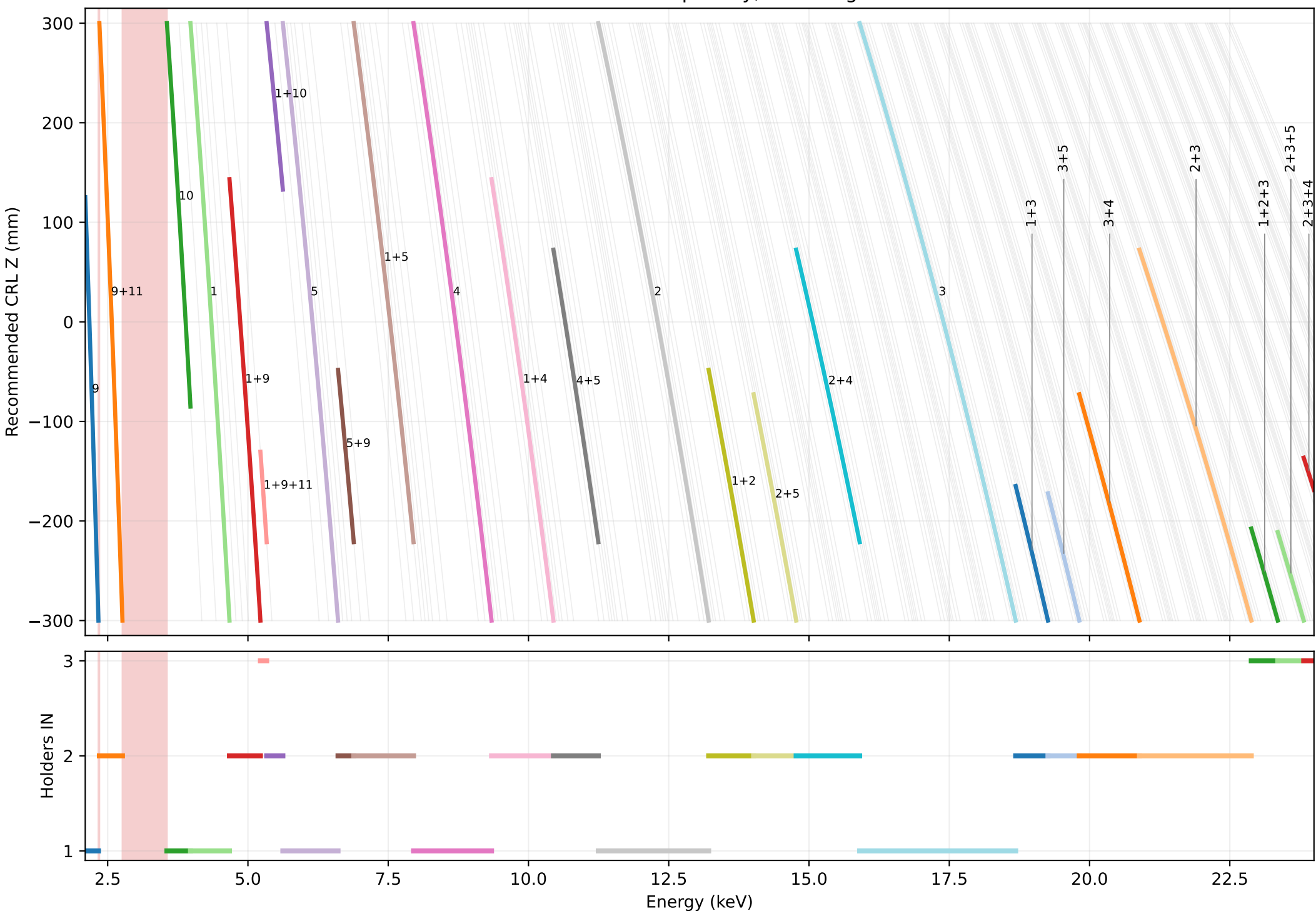


$p=10.800$ m; $q=1.300$ m; required $f=1160.3$ mm
Holder 3 example: 15.902 keV



Positive CRL Z increases source-to-lens distance p and decreases q .
Focusing condition: $1/f = 1/p + 1/q$; SSA and sample stay fixed.
Longitudinal scale is physical; transverse ray heights are arbitrary.
Rays illustrate focusing geometry, not predicted beam size or aperture.

Fewest-holder selection; labels are holder numbers; gray = alternatives
Minimum holder count takes priority, including at travel limits



Recommended energy bands

Energy (keV)	Holders IN	N	Z start -> Z end (mm)
2.100- 2.335	9	1	125.3 -> -300.0
2.335- 2.352	UNREACHABLE		
2.352- 2.763	9+11	2	300.0 -> -300.0
2.763- 3.556	UNREACHABLE		
3.556- 3.976	10	8	300.0 -> -85.1
3.976- 4.671	1	1	300.0 -> -300.0
4.671- 5.222	1+9	2	143.3 -> -300.0
5.222- 5.334	1+9+11	3	-130.5 -> -221.2
5.334- 5.622	1+10	9	300.0 -> 132.8
5.622- 6.606	5	2	300.0 -> -300.0
6.606- 6.886	5+9	3	-48.2 -> -221.2
6.886- 7.951	1+5	3	300.0 -> -221.2
7.951- 9.342	4	4	300.0 -> -300.0
9.342- 10.444	1+4	5	143.3 -> -300.0
10.444- 11.245	4+5	6	72.4 -> -221.2
11.245- 13.211	2	8	300.0 -> -300.0
13.211- 14.013	1+2	9	-48.2 -> -300.0
14.013- 14.771	2+5	10	-72.7 -> -300.0
14.771- 15.902	2+4	12	72.4 -> -221.2
15.902- 18.683	3	16	300.0 -> -300.0
18.683- 19.258	1+3	17	-164.9 -> -300.0
19.258- 19.817	3+5	18	-172.2 -> -300.0
19.817- 20.889	3+4	20	-72.7 -> -300.0
20.889- 22.883	2+3	24	72.4 -> -300.0
22.883- 23.354	1+2+3	25	-207.6 -> -300.0
23.354- 23.817	2+3+5	26	-211.1 -> -300.0
23.817- 24.000	2+3+4	28	-136.4 -> -168.8

Endpoints rounded; use selection_bands.csv for precision.

At shared endpoints both adjacent choices can be reachable.

Use recommend() at the exact energy, not interpolation across a holder change. N is the total number of individual lenses.

All 255 known combinations evaluated; fewest holders first, then fewest elements, then holder IDs. Travel limits included.